STM - Exercise 1

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Exercises: let’s post your results on the Padlet for the discussion
Post also your doubts!

https://padlet.com/polito/exercise-1-team-2-ww3fplotu45s5vbm

Exercise 1

Use figure 1 to:

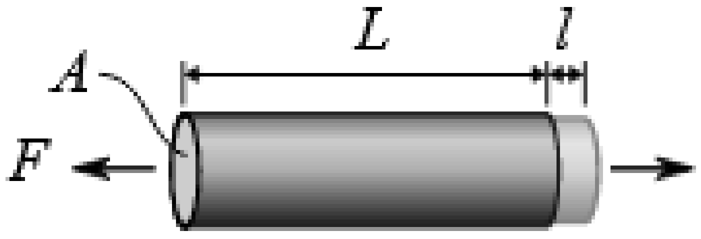
a) define which test and which class of materials is referred to and why;

b) calculate elastic modulus,

c) Calculate the yield strength (0.2 % offset) and the tensile strength,

d) Calculate the elongation at fracture for a 50 mm long specimen (<u>write on curves</u> data you use).

e) Calculate the length of the specimen when you keep it loaded in A.



$$\sigma = \frac{F}{A}$$

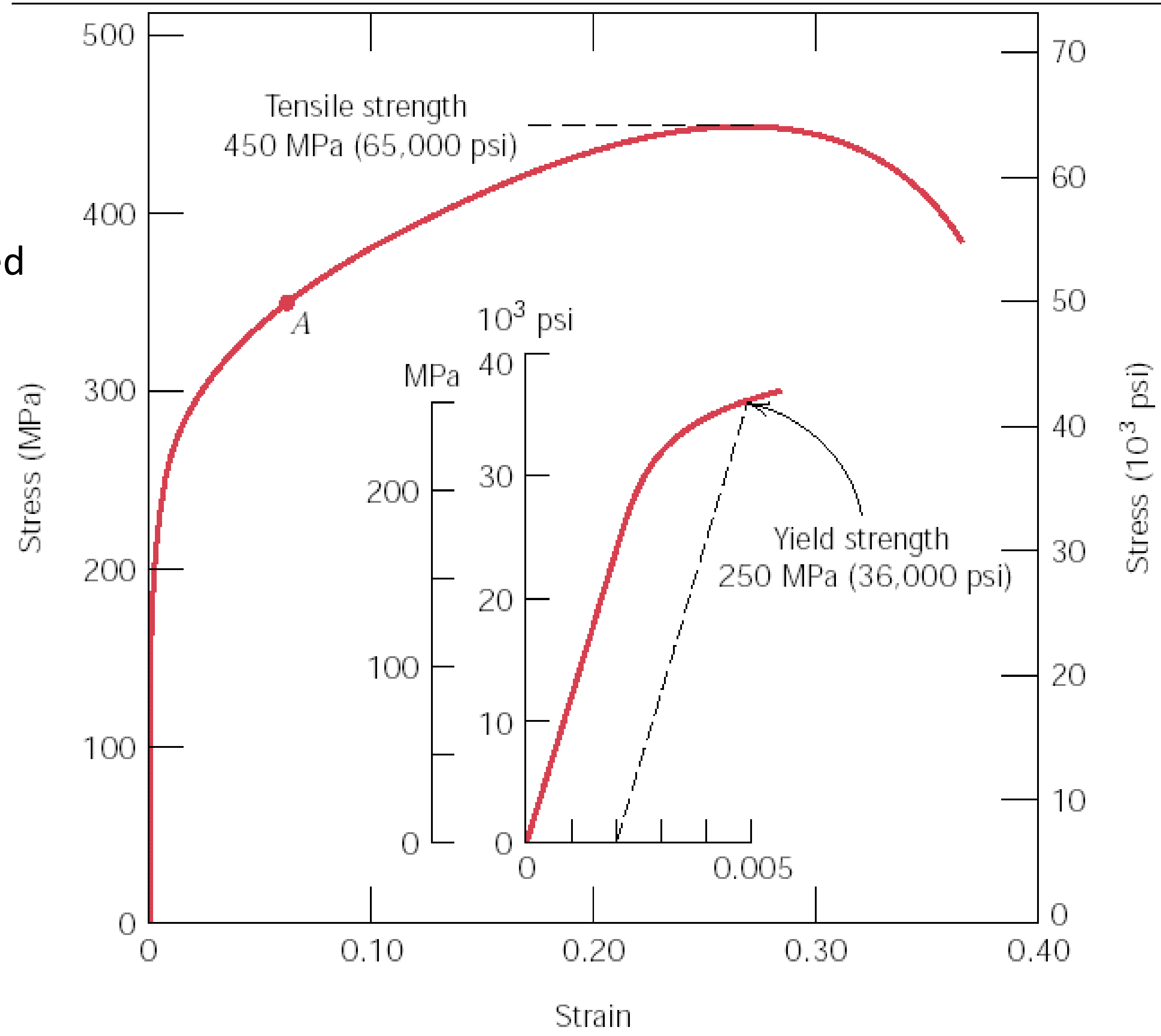
where

- σ stress [Pa]
- F applied force [N]
- A cross-sectional area [m^2]

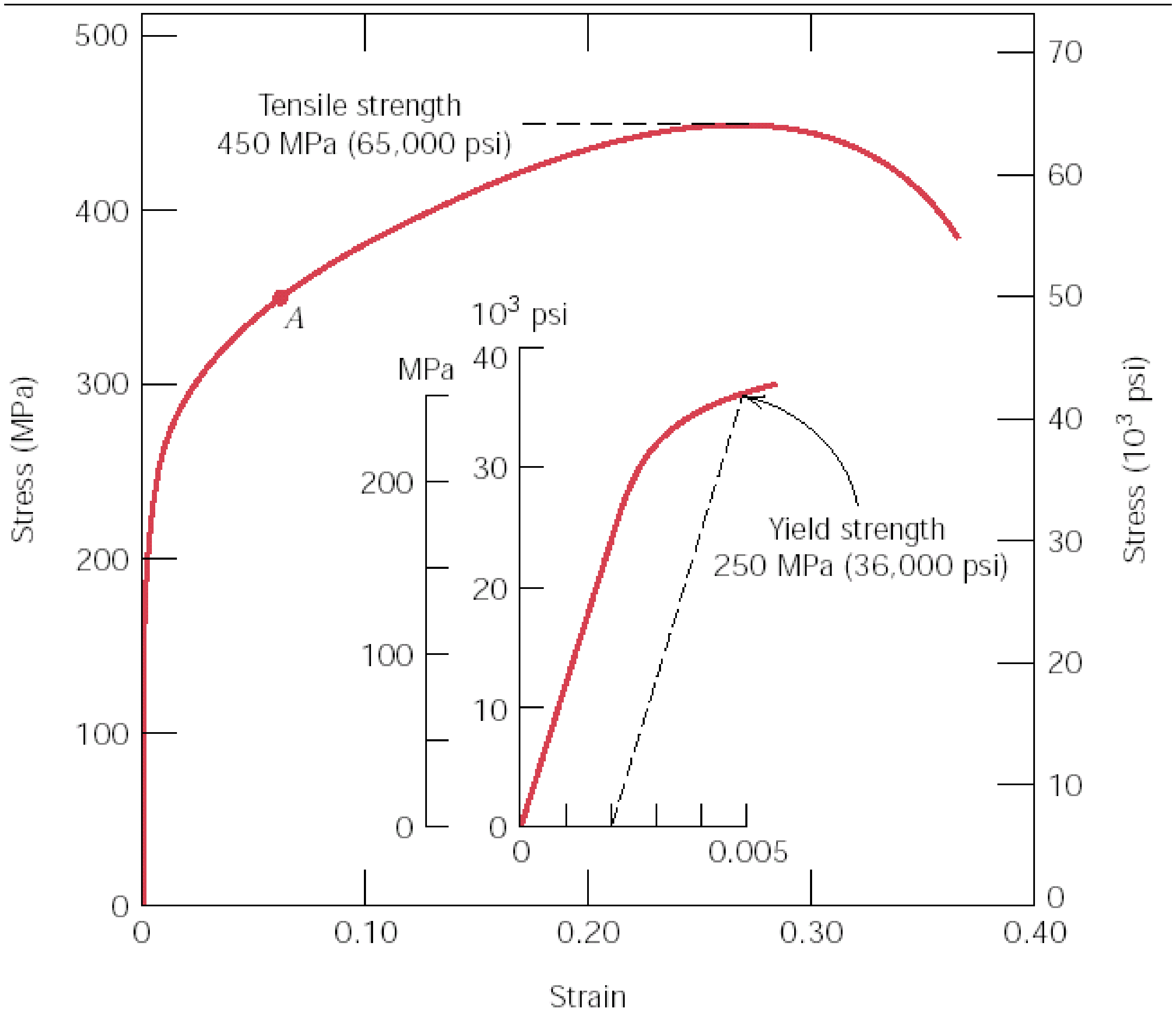
$$\varepsilon = \frac{\Delta L}{L_0}$$

where

- ε strain
- ΔL total elongation [m]
- L_0 original length [m]

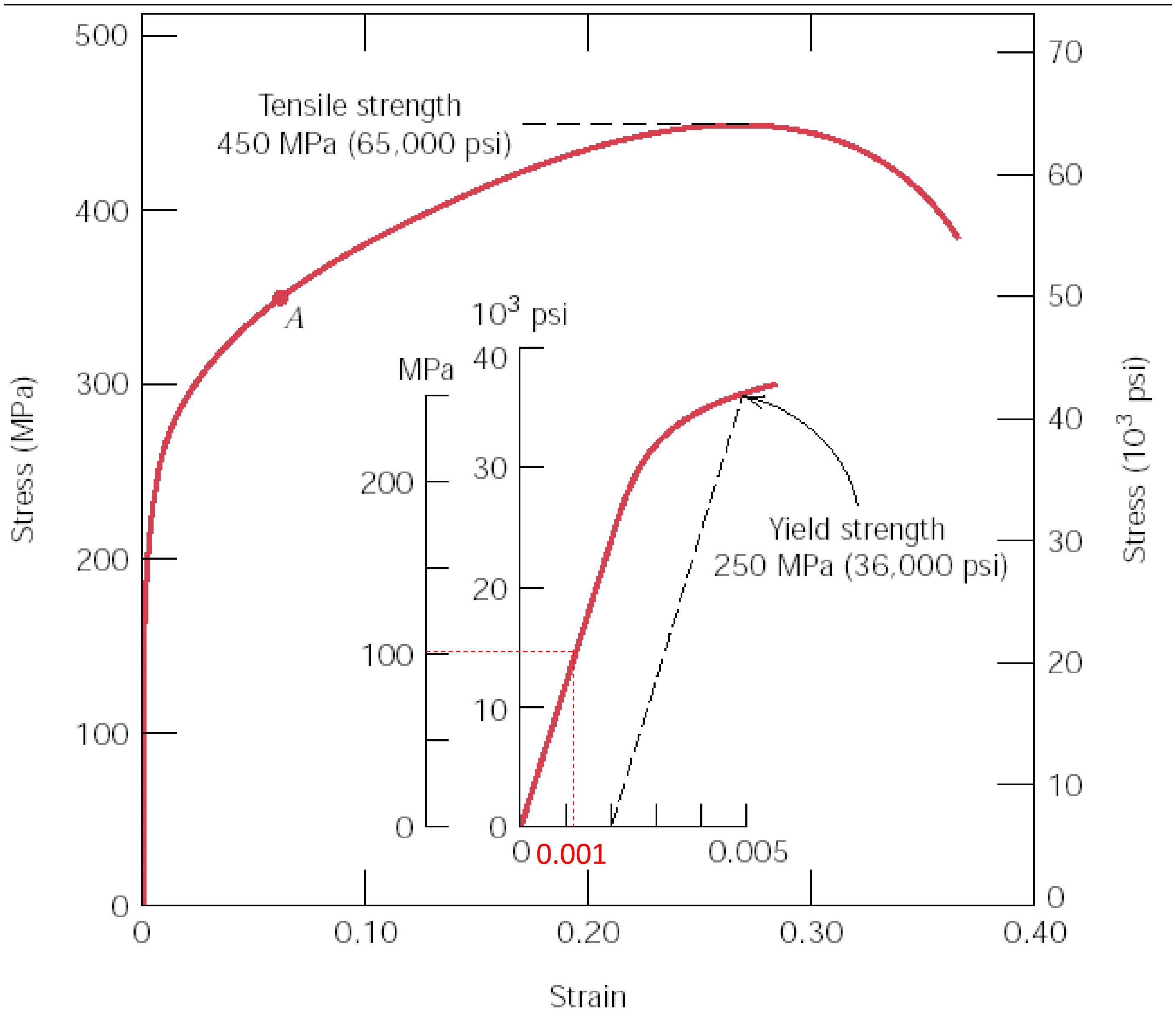


Exercise 1a: Use figure 1 to define which test and which class of materials is referred to and why



Typical stress-strain curve from tensile test
Ductile behavior and stress-strain values typical of metallic materials

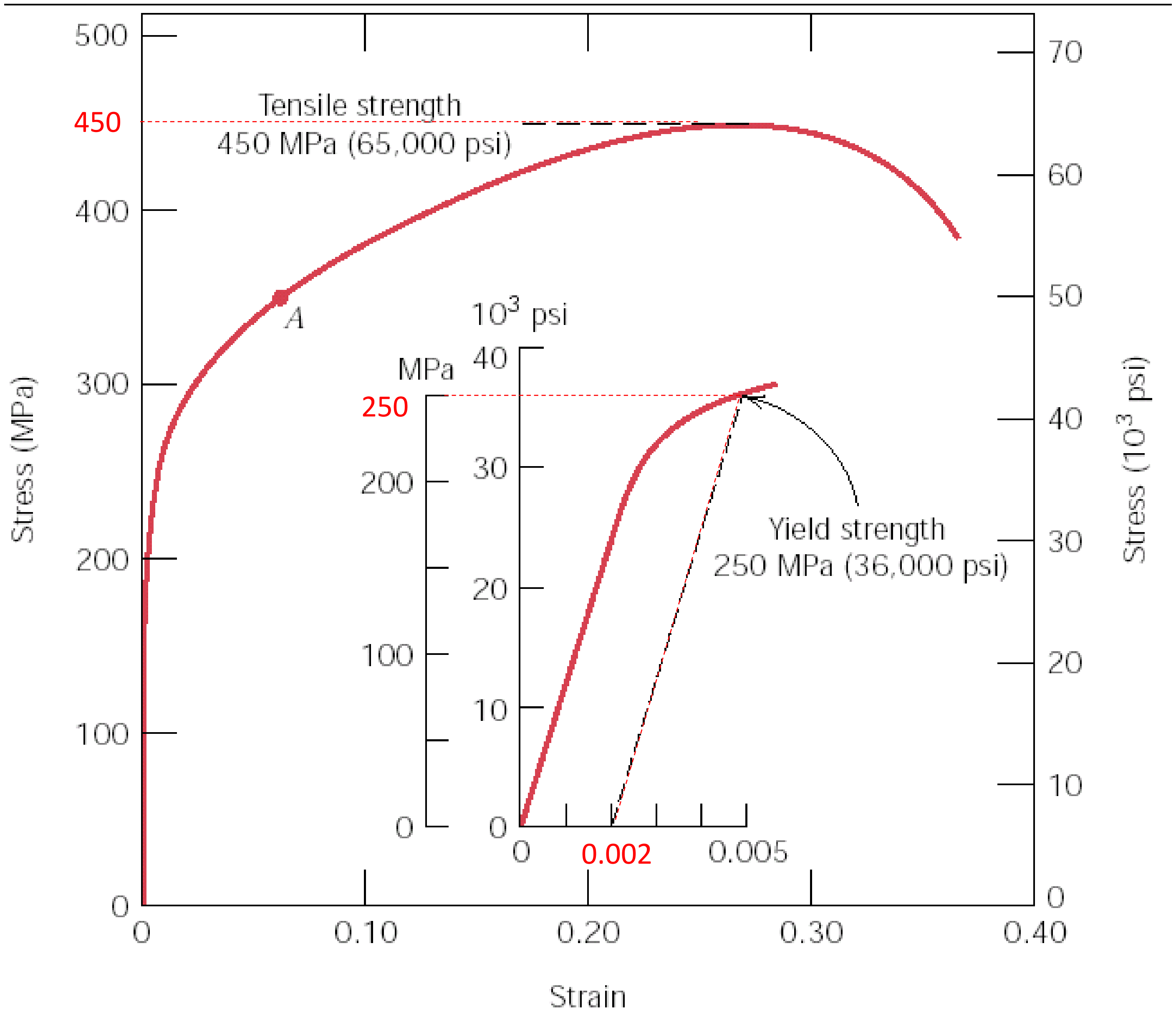
Exercise 1b: Calculate elastic modulus



$$E = \frac{\sigma}{\varepsilon}$$

E= 100MPa/0.001=
100000 Mpa= 100 GPa

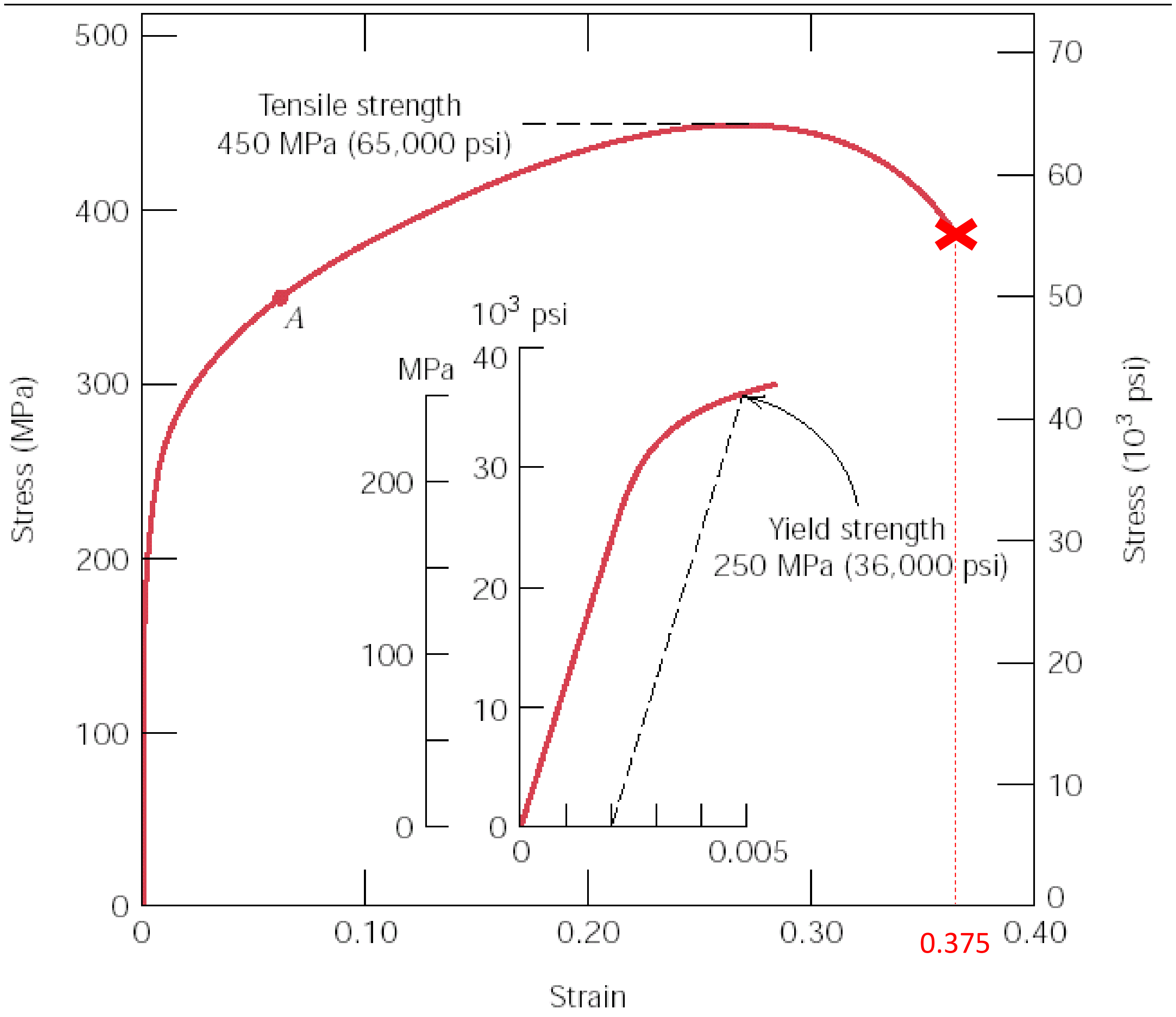
Exercise 1c: calculate yield strength (0.2 % offset) and tensile strength



yield strength (0,2 % offset) =250 MPa *from the graph*

tensile strength =450 MPa *maximum of the curve, from the graph*

Exercise 1d: Calculate elongation at fracture for a 50 mm long specimen (<u>write on curves</u> data you use).

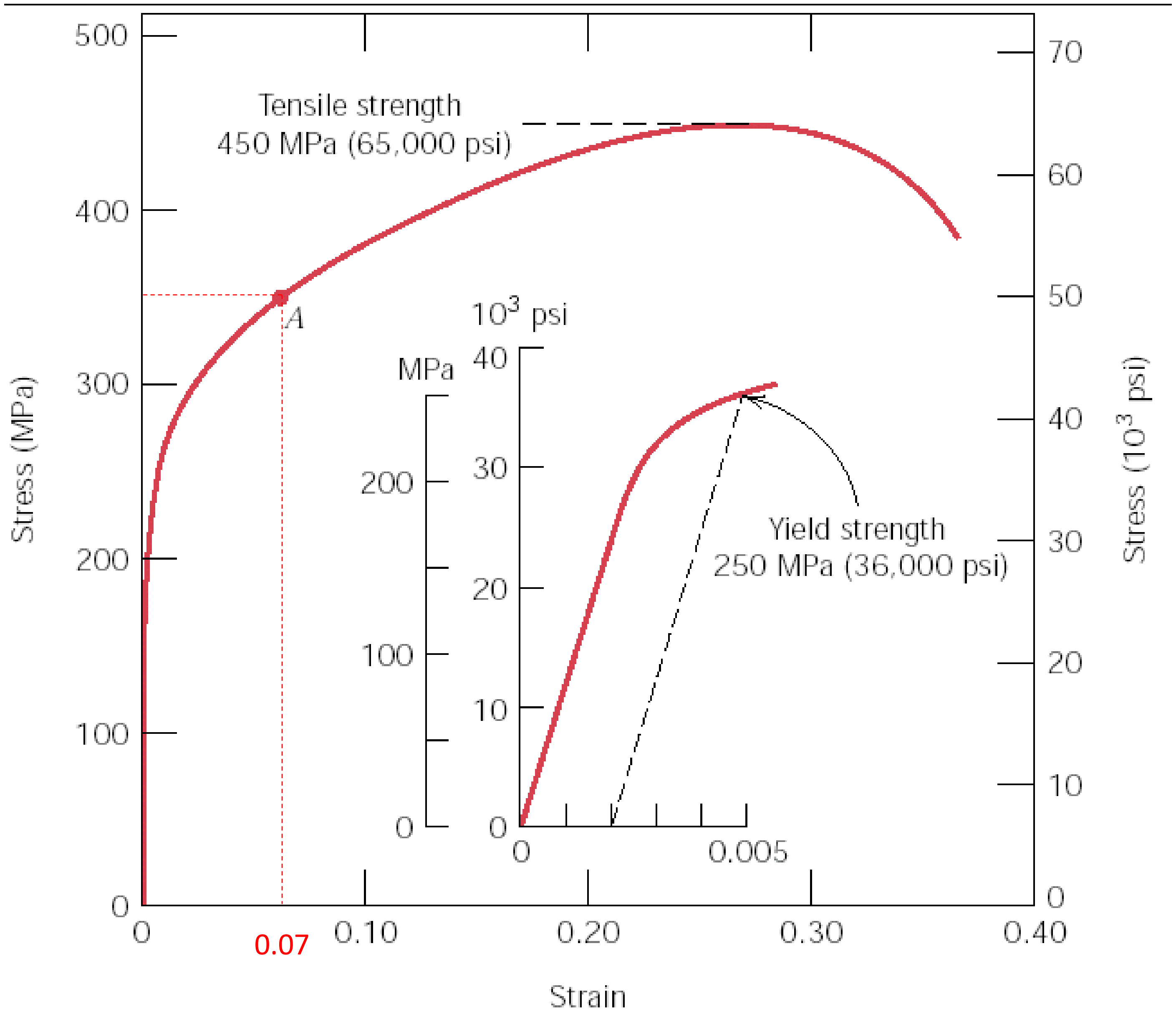


ε_F=0.375

$\varepsilon_F = \Delta l_F / l_0$

$\Delta l_F = \varepsilon_F * l_0 = 0.375*50 = 18.75$ mm

Exercise 1e- Calculate the length of the specimen when you keep it loaded in A.



$\boldsymbol{\varepsilon_A}$=0.07

$\boldsymbol{\varepsilon_A}$=$\Delta l_A/l_0$

Δl_A= $\boldsymbol{\varepsilon_A}$ ***** l_0 = 0.07*50= 3.5 mm

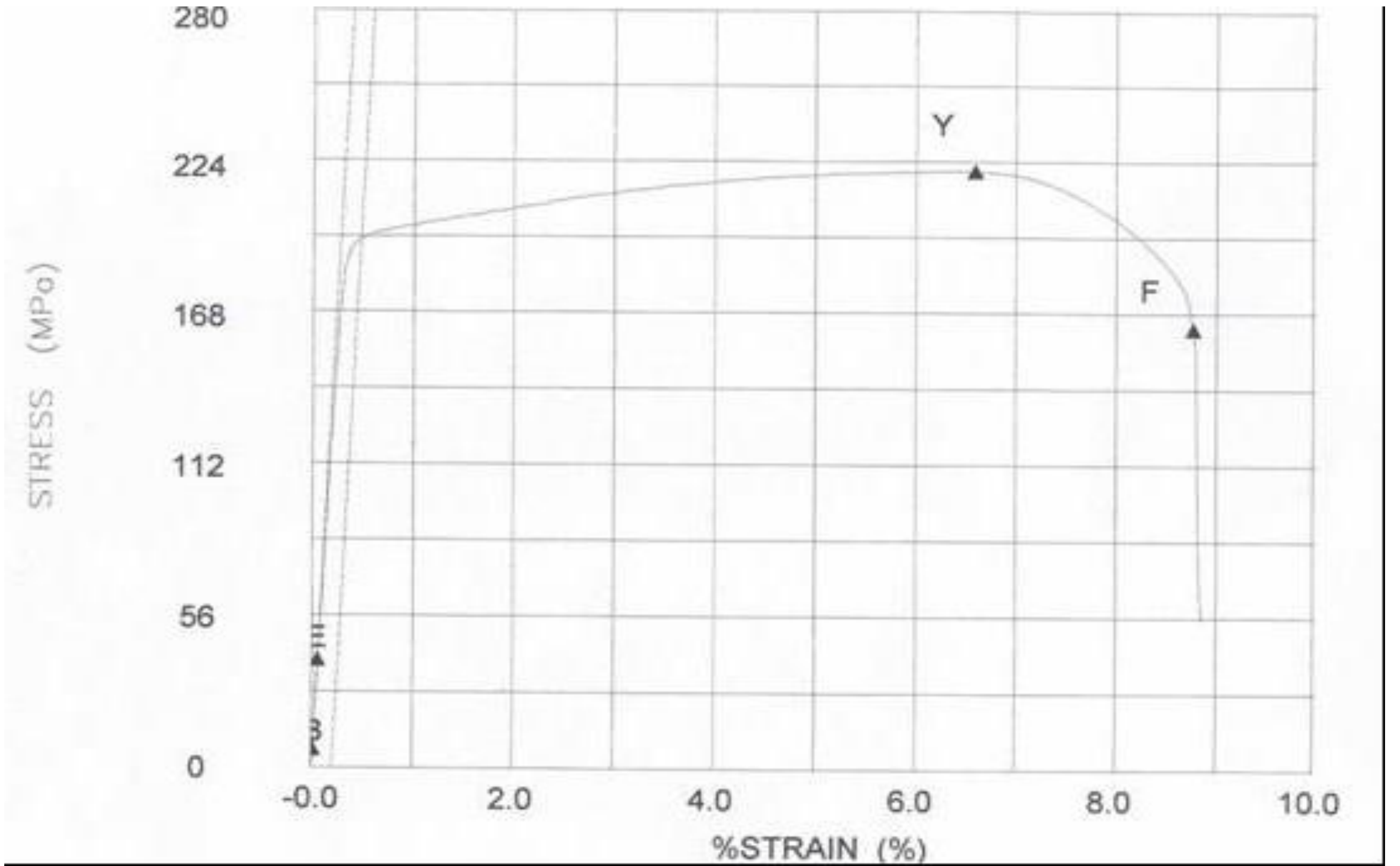
l_A=l_0+ Δl_A= 50+3.5=53.5 mm

<u>If you keep the sample loaded no recovery of elastic deformation occurs</u>

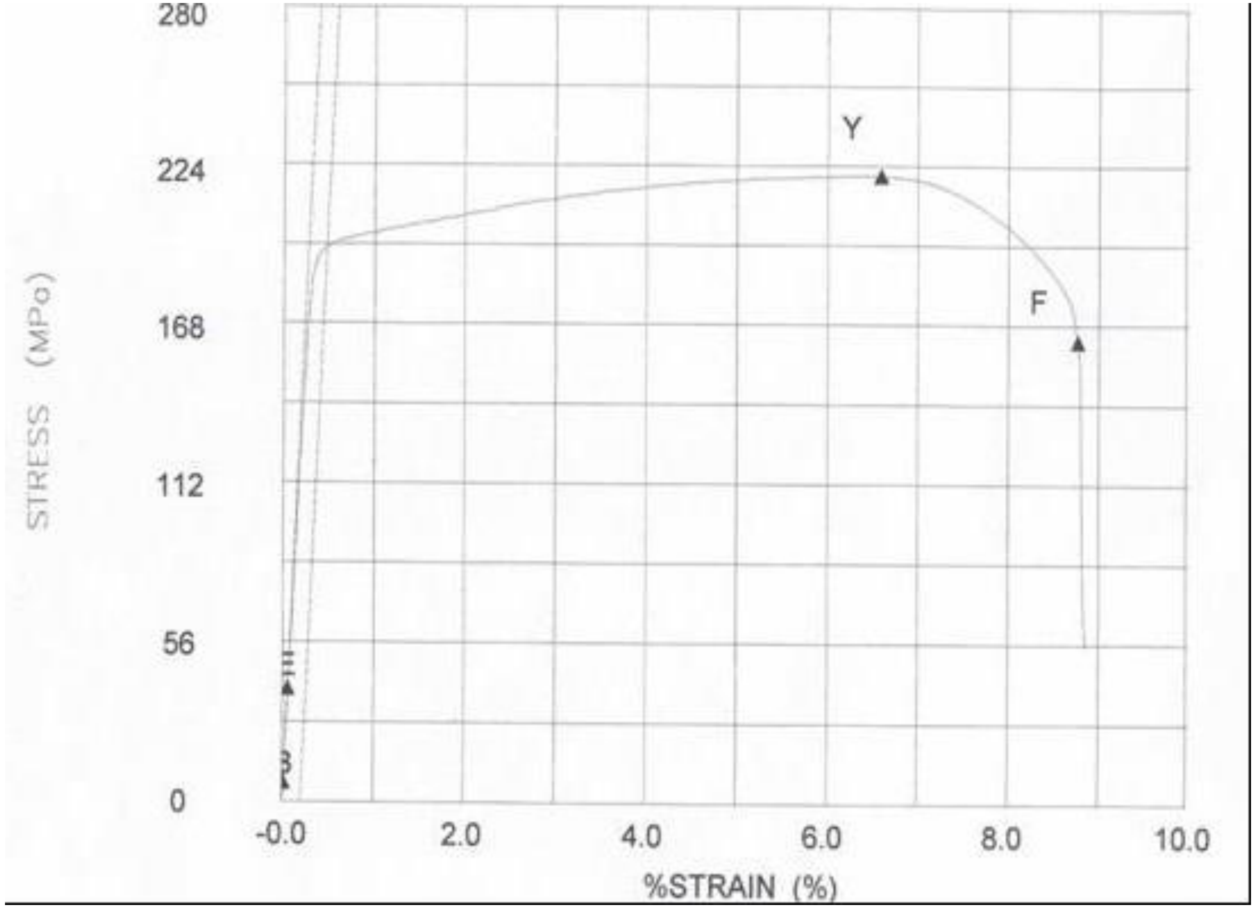
Exercise 2: Use the graph below to define

a) which test and which class of materials is referred to and why;
b) to calculate elastic modulus,
c) To calculate the yield strength (0,2 % strain) and the tensile strength,
d) To calculate the elongation at fracture for a 50 mm long specimen;
e) To calculate the elongation of this sample when loaded at 200 MPa, then unloaded.

(write on graph stress and strain
you use to calculate).

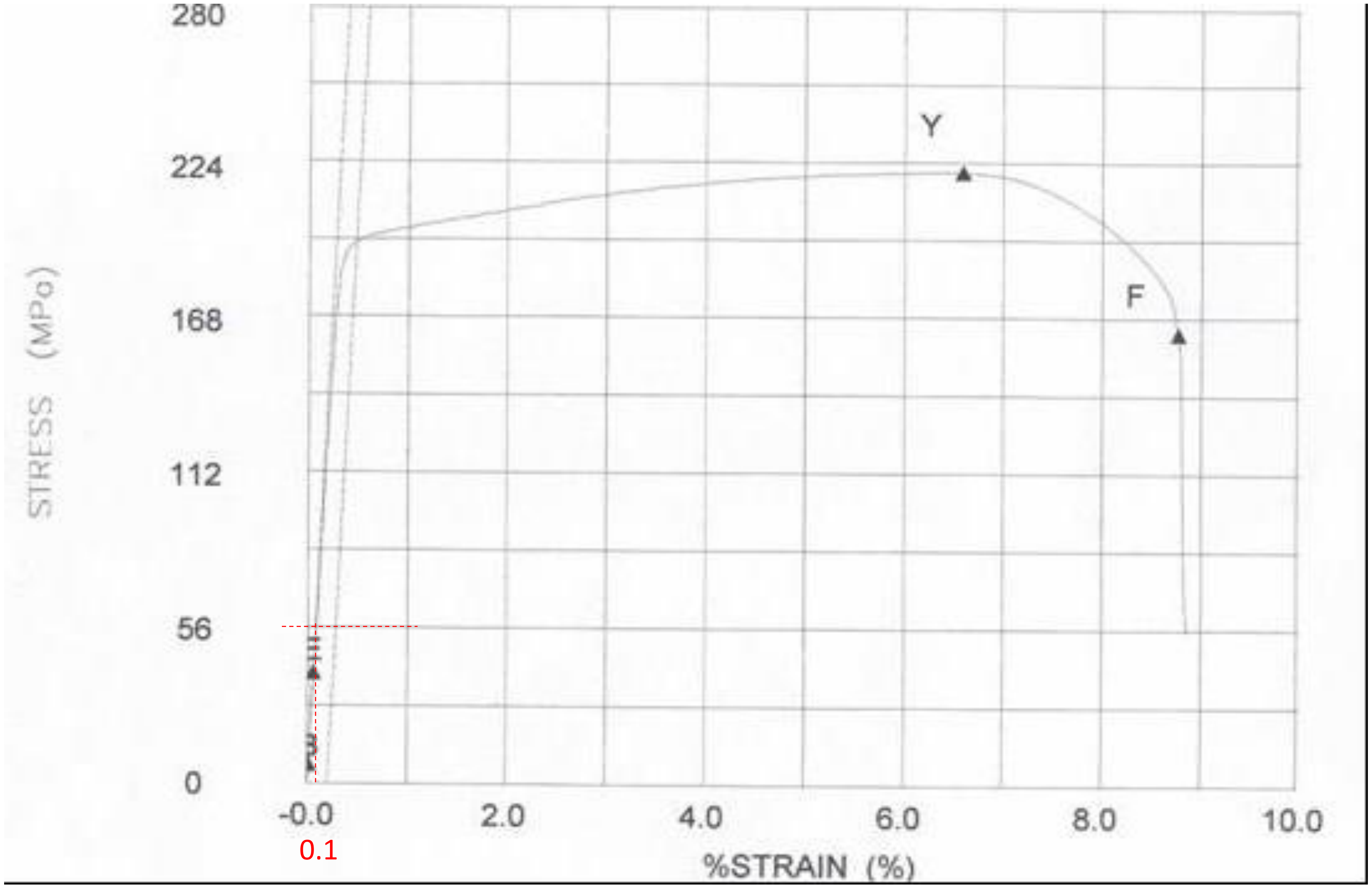


Exercise 2a: Define which test and which class of materials is referred to and why



Typical stress-strain curve from tensile test
Ductile behavior and stress-strain values typical of metallic materials

Exercise 2b: calculate elastic modulus

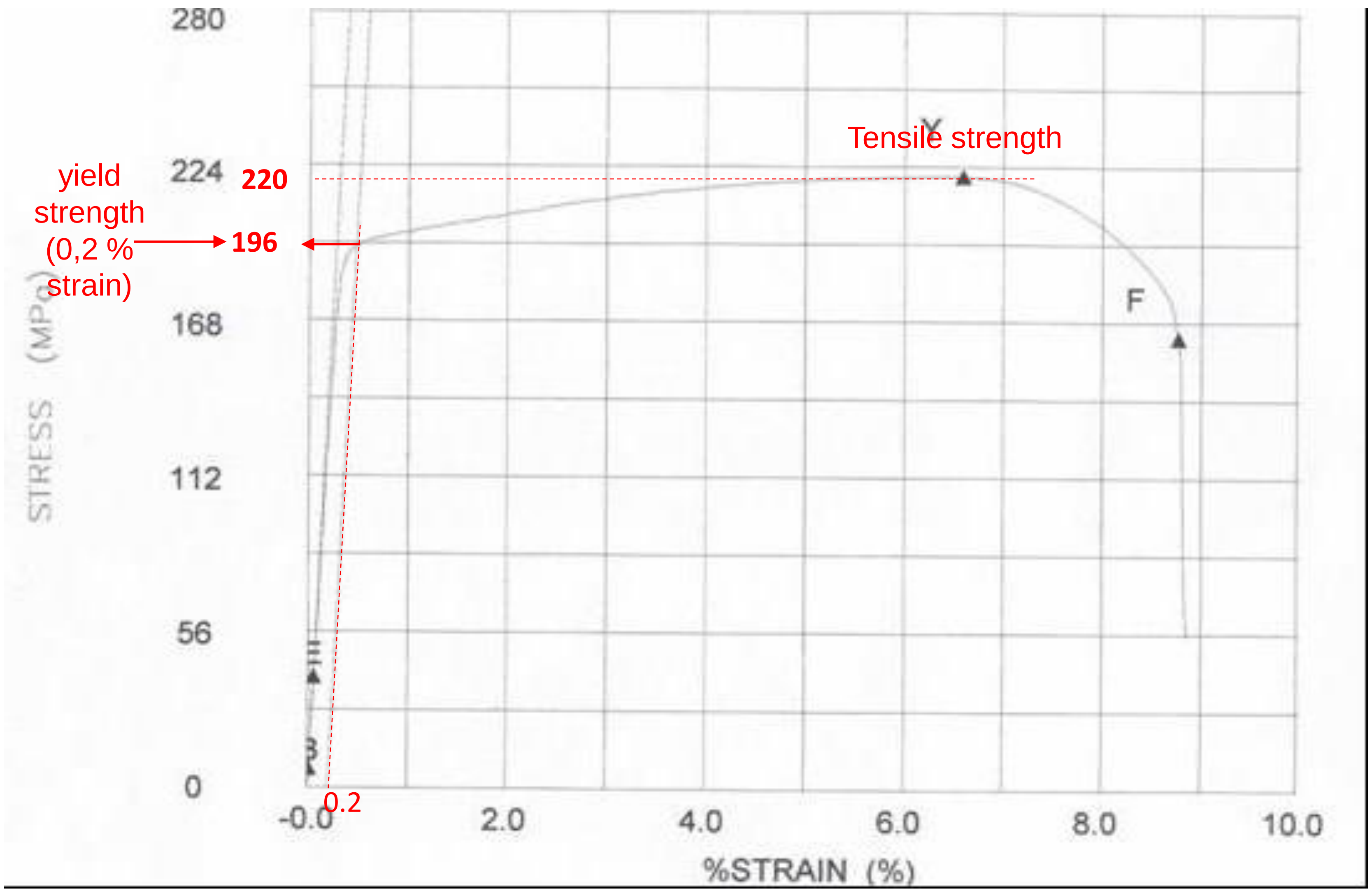


ε=0.1%=0.001

E=σ/ε=

56/0.001=56000MPa=56GPa

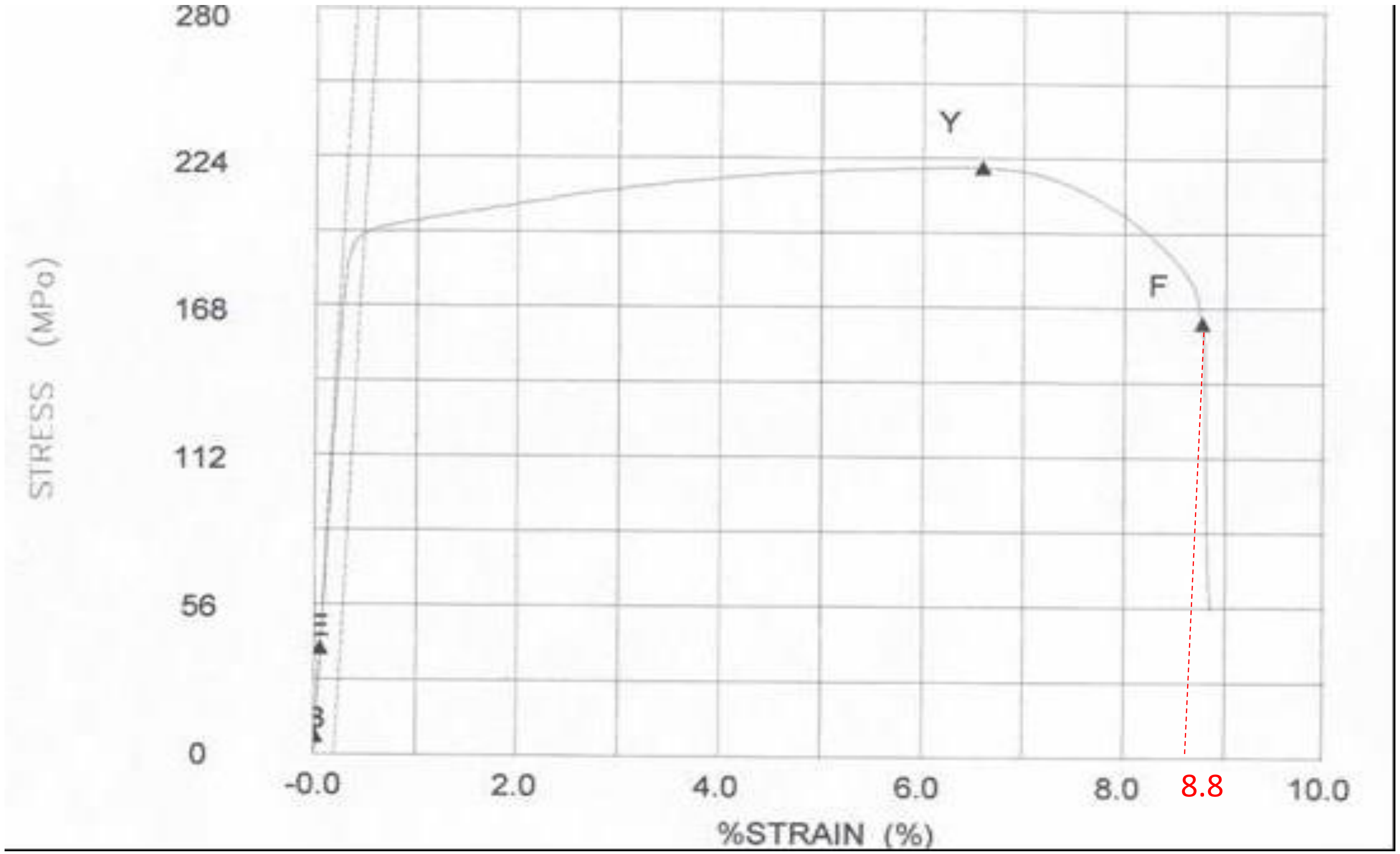
Exercise 2c: calculate yield strength (0,2 % strain) and tensile strength



yield strength (0,2 % strain) =196 MPa *from the graph*

tensile strength =220 MPa *maximum of the curve, from the graph*

Exercise 2d: Calculate elongation at fracture for a 50 mm long specimen

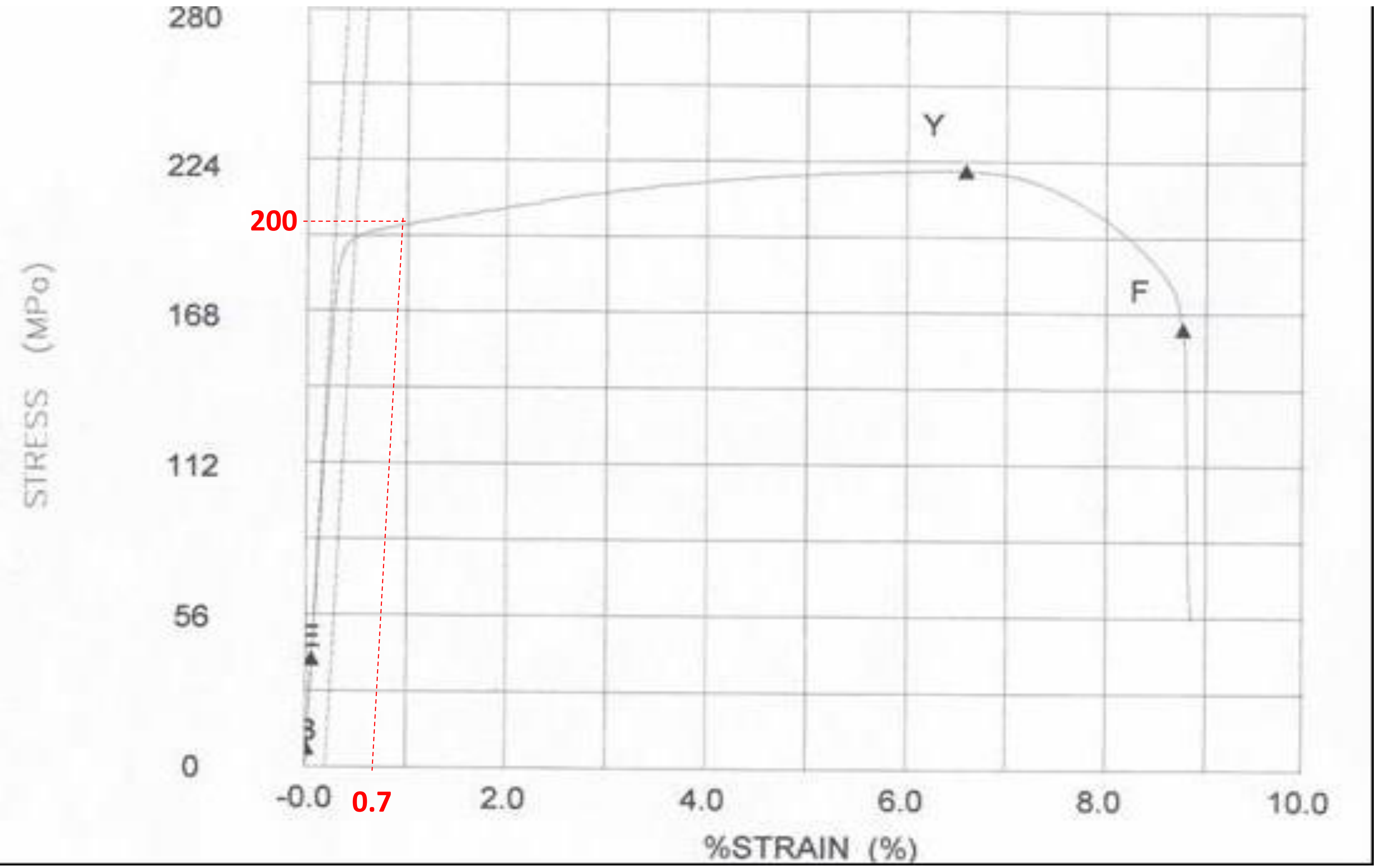


ε_F=8.8%=0.088

$\varepsilon_F=\Delta l_F/l_0$

$\Delta l_F = \varepsilon_F * l_0 = 0.088*50 = 4.4$ mm

Exercise 2e: Calculate the elongation of this sample when loaded at 200 MPa, then unloaded. (write on graph stress and strain you use to calculate).

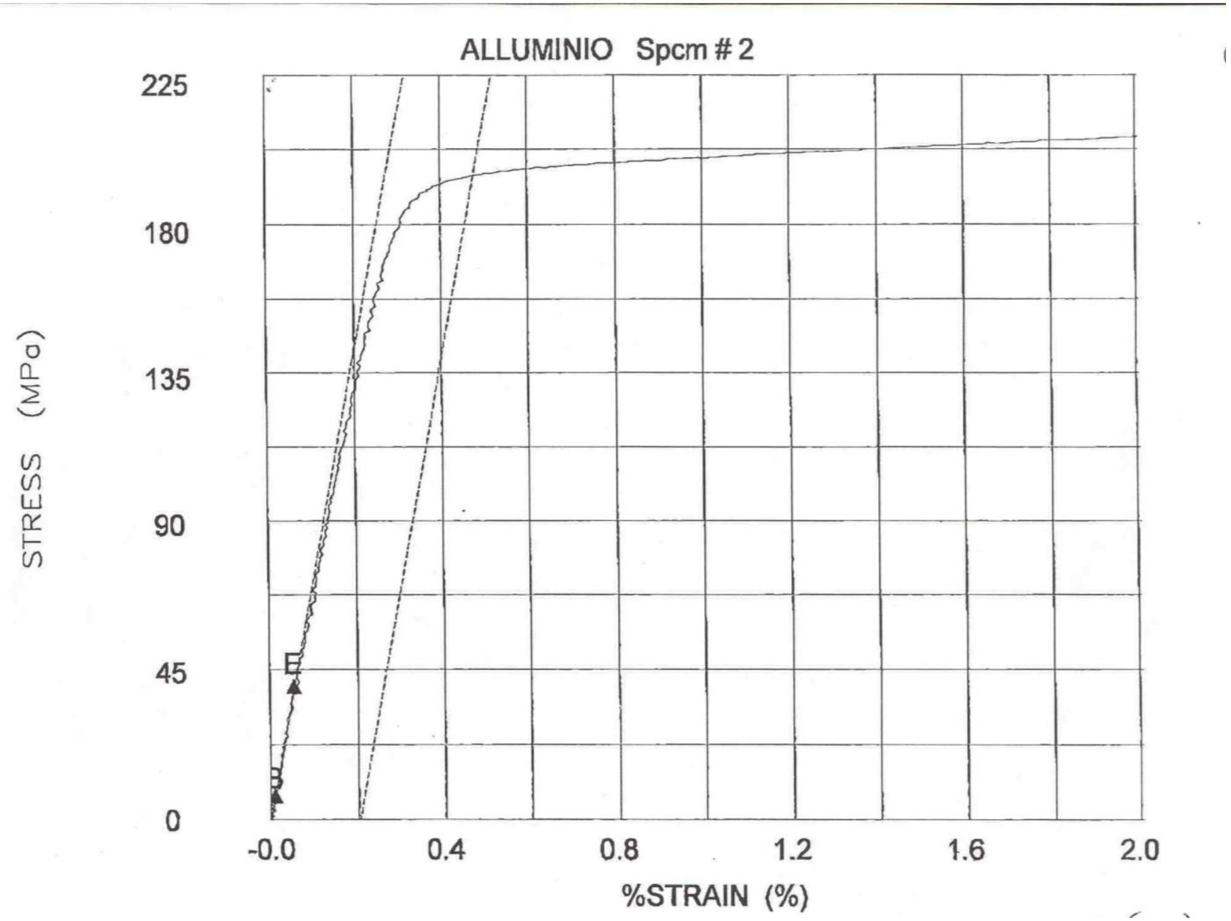


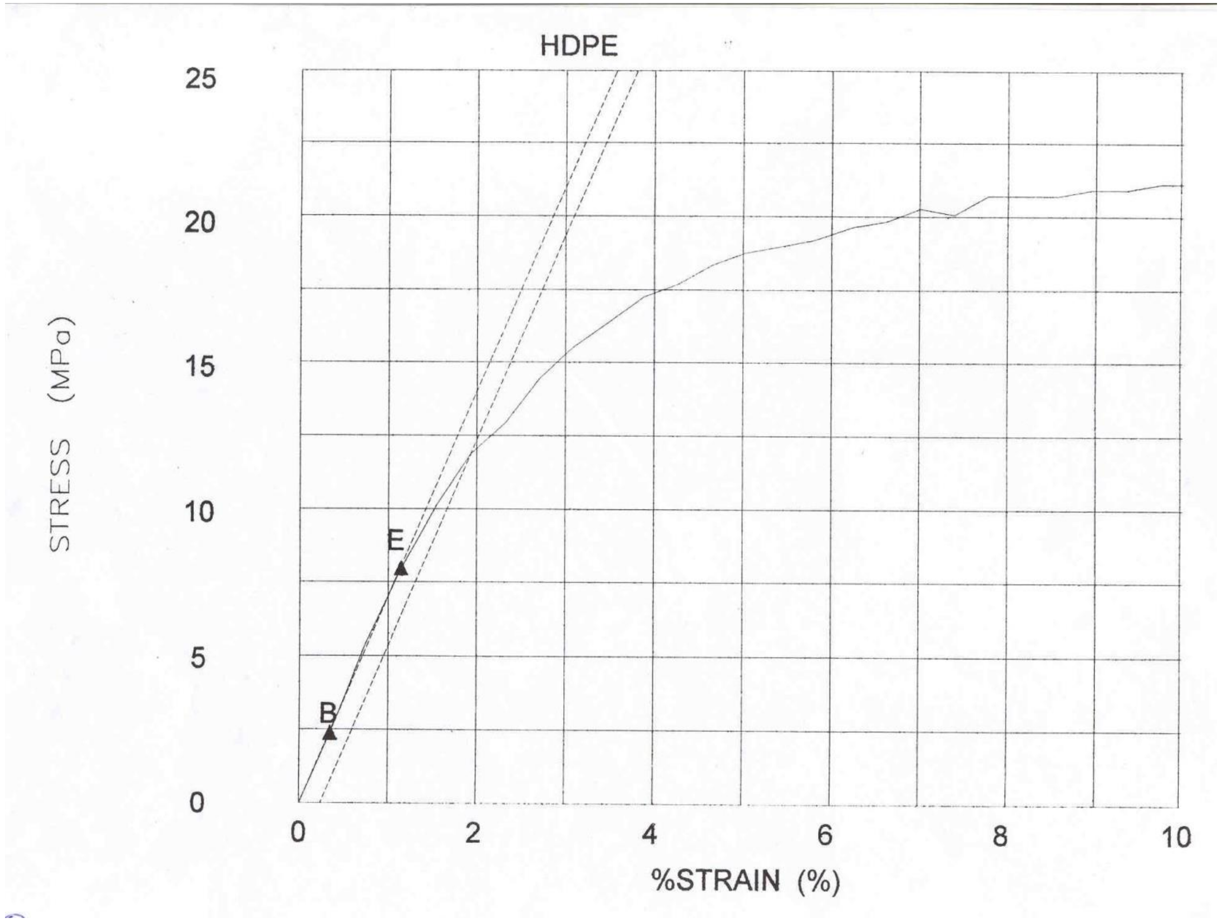
Loading and unloading → elastic deformation recovered, only permanent deformation present

$\varepsilon = 0.7\% = 0.007$

$\Delta l = \varepsilon * l_0 = 0.007 * 50 = 0.35 mm$

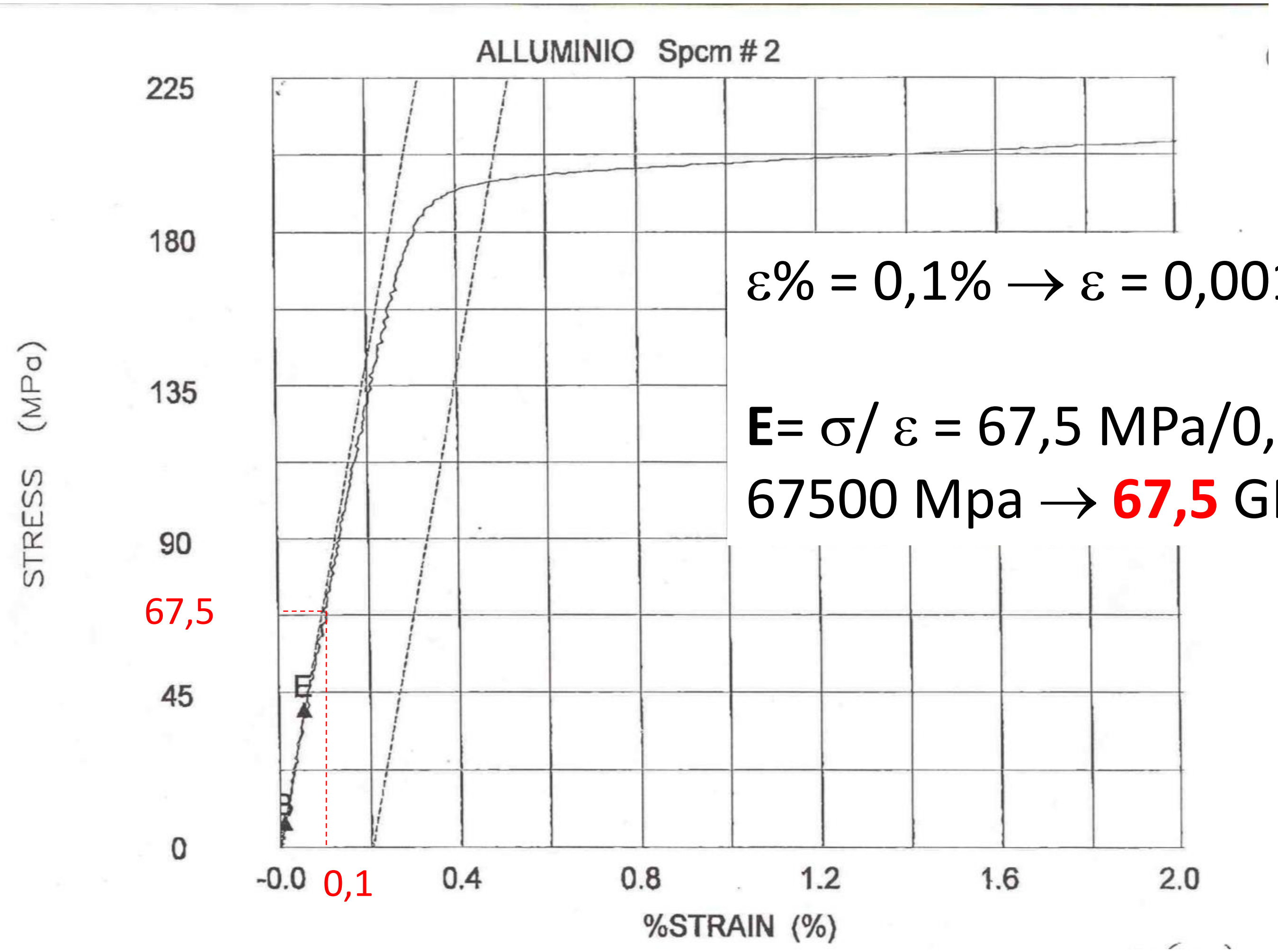
Exercise 3: Determine the Elastic Modulus of Aluminum and High Density Polyethylene (HDPE) from the graphs below. Comment the obtained results.





Exercise 3 – case 1: Aluminum

Elastic modulus (E) = slope of the linear (elastic) tract of the σ-ε curve

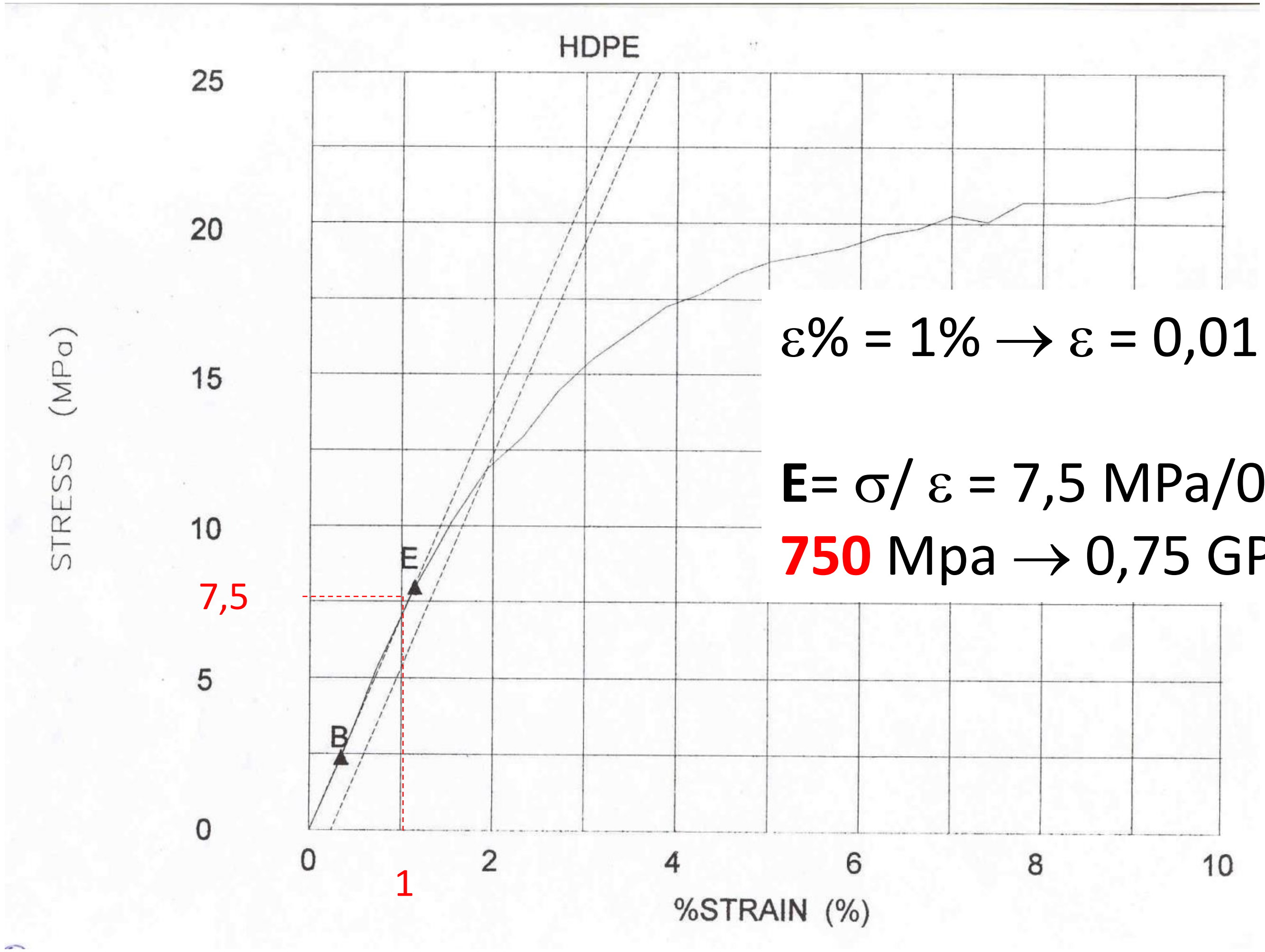


$\varepsilon\% = 0{,}1\% \rightarrow \varepsilon = 0{,}001$

E$= \sigma / \varepsilon = 67{,}5$ MPa$/0{,}001 =$ 67500 Mpa $\rightarrow$ **67,5** GPa

Exercise 3 – case 2: HDPE

Elastic modulus (E) = slope of the linear (elastic) tract of the σ-ε curve



$\varepsilon\% = 1\% \rightarrow \varepsilon = 0{,}01$

E$= \sigma/\varepsilon = 7{,}5$ MPa$/0{,}01=$
750 Mpa $\rightarrow$ 0,75 GPa

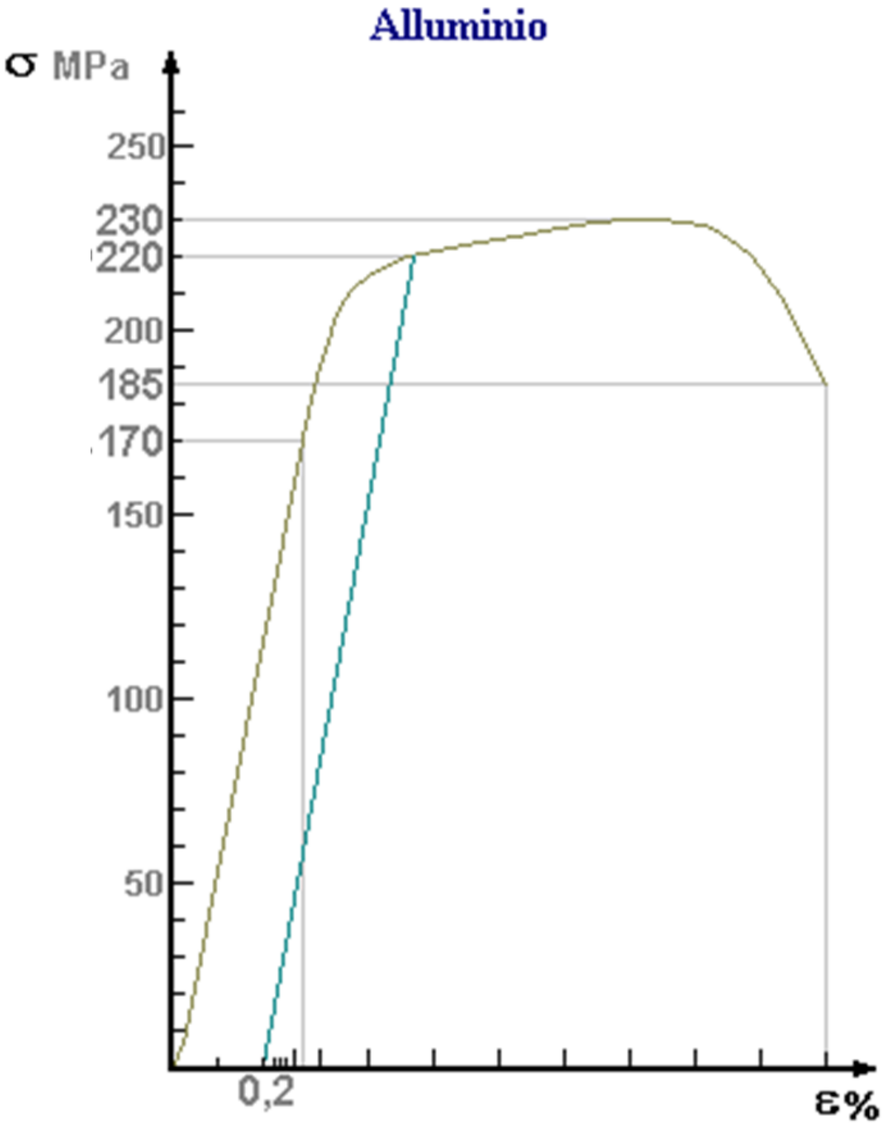
Exercise 3 comparison Aluminum – HDPE

EAl=67.5 Gpa
E HDPE= 0.75 Gpa

1 order of magnitude difference

Elastic modulus depends on the bonding strenght: metallic bond vs van der Waals interactions

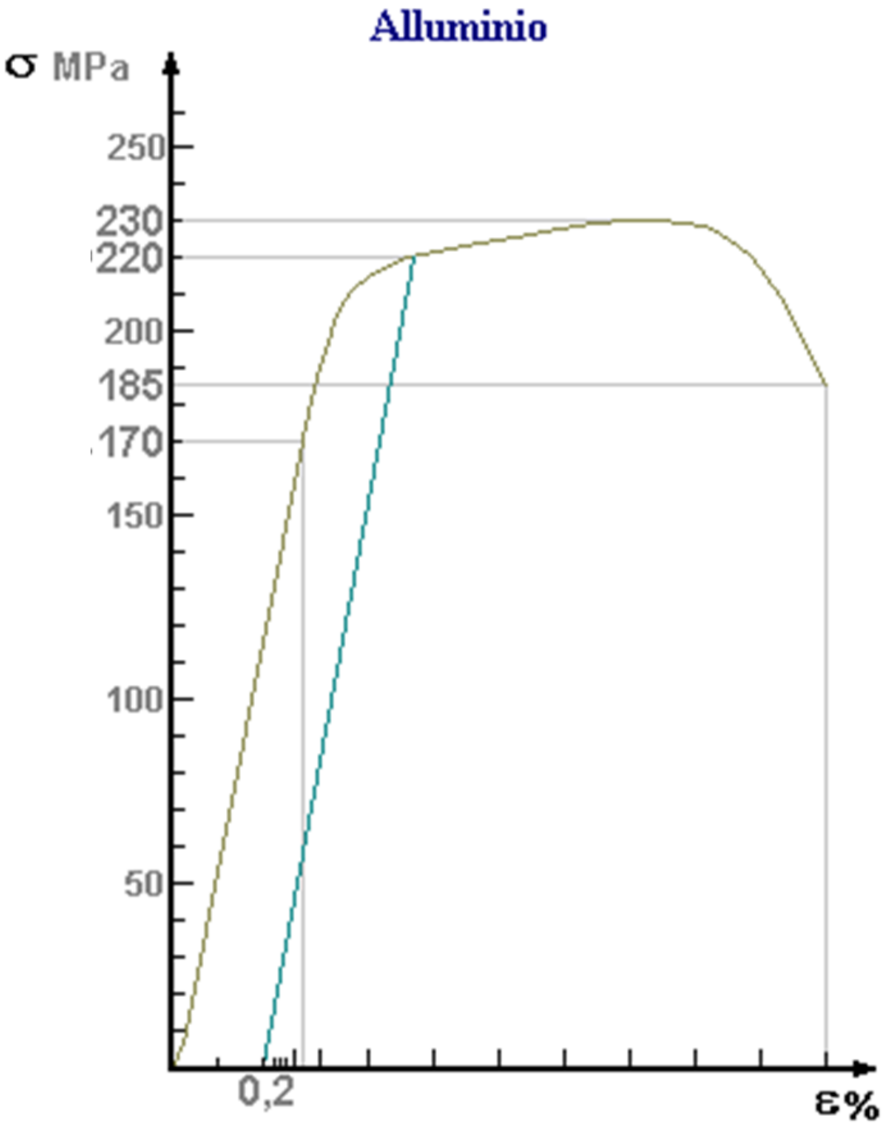
Quiz 1



Consider the σ-ε curve reported in the figure.
The yield strength (0.2 % offset) is:

a) 230 Mpa
b) 220 Mpa
c) 185 MPa

Quiz 1



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a) 230 Mpa
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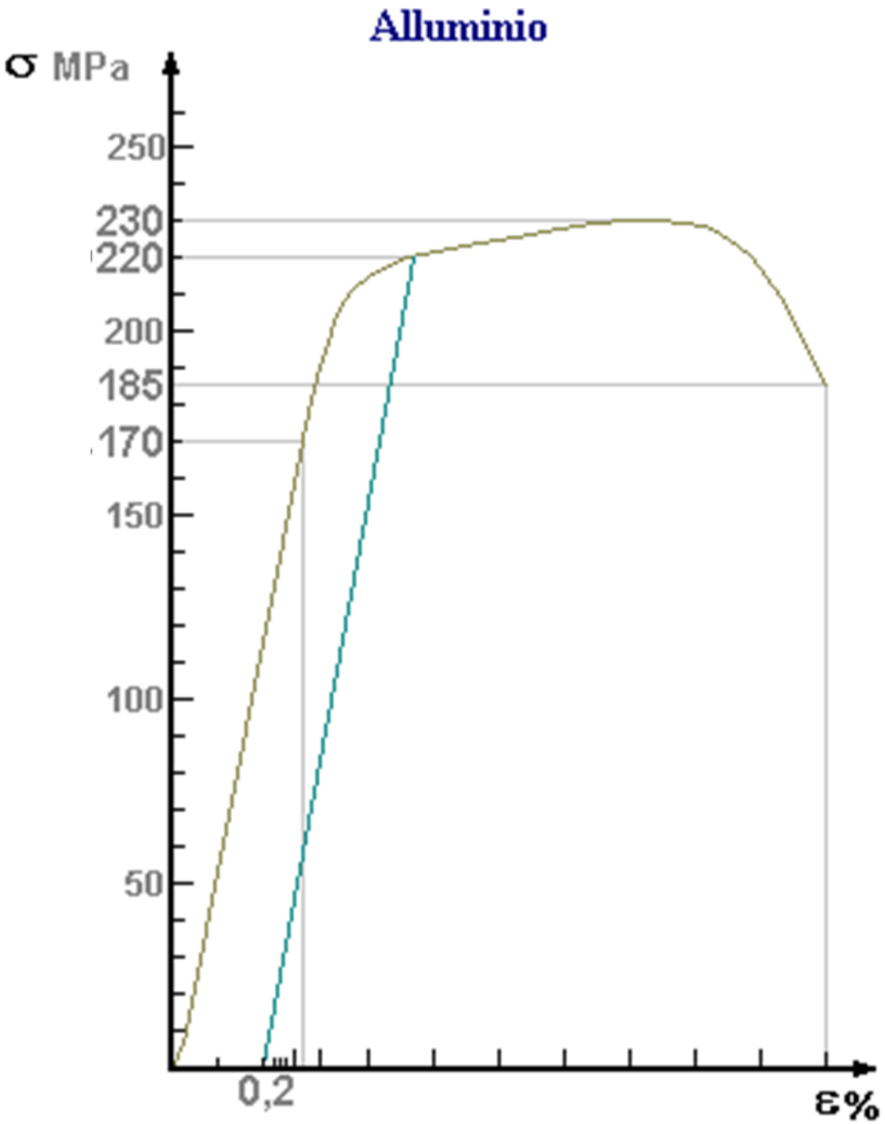
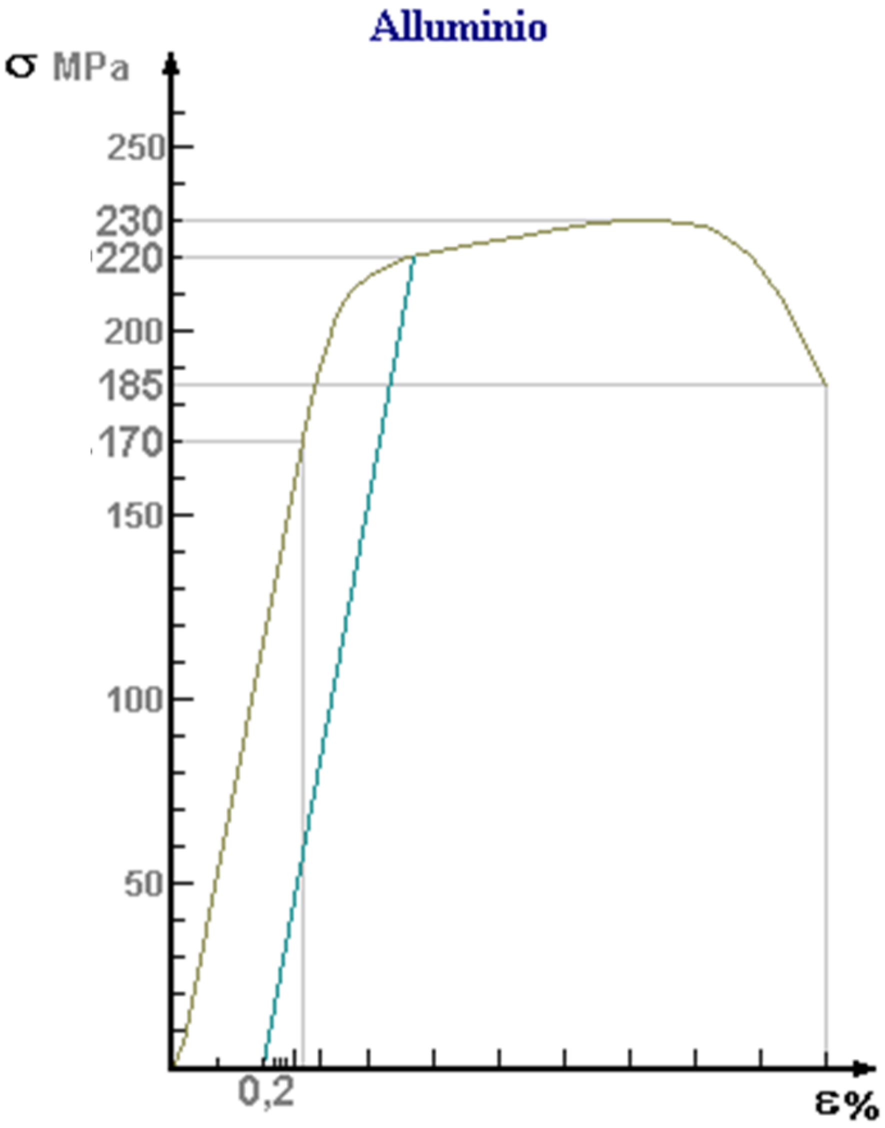
Quiz 2



The σ-ε curve in figure comes from the tensile test of an aluminum specimen.
Select the right sentence (only one):

a) The specimen underwent plastic deformation
b) The specimen underwent fragile fracture
c) The specimen did not have permanent deformations

Quiz 2



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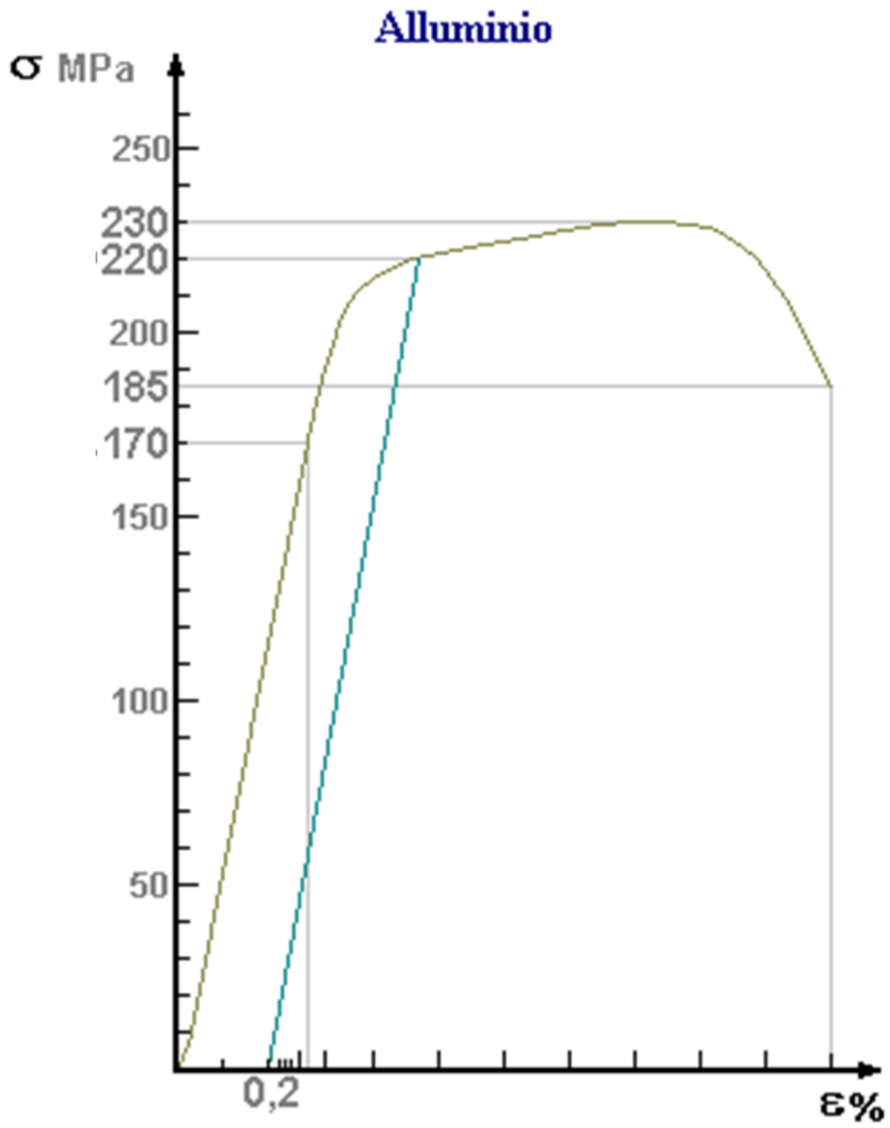


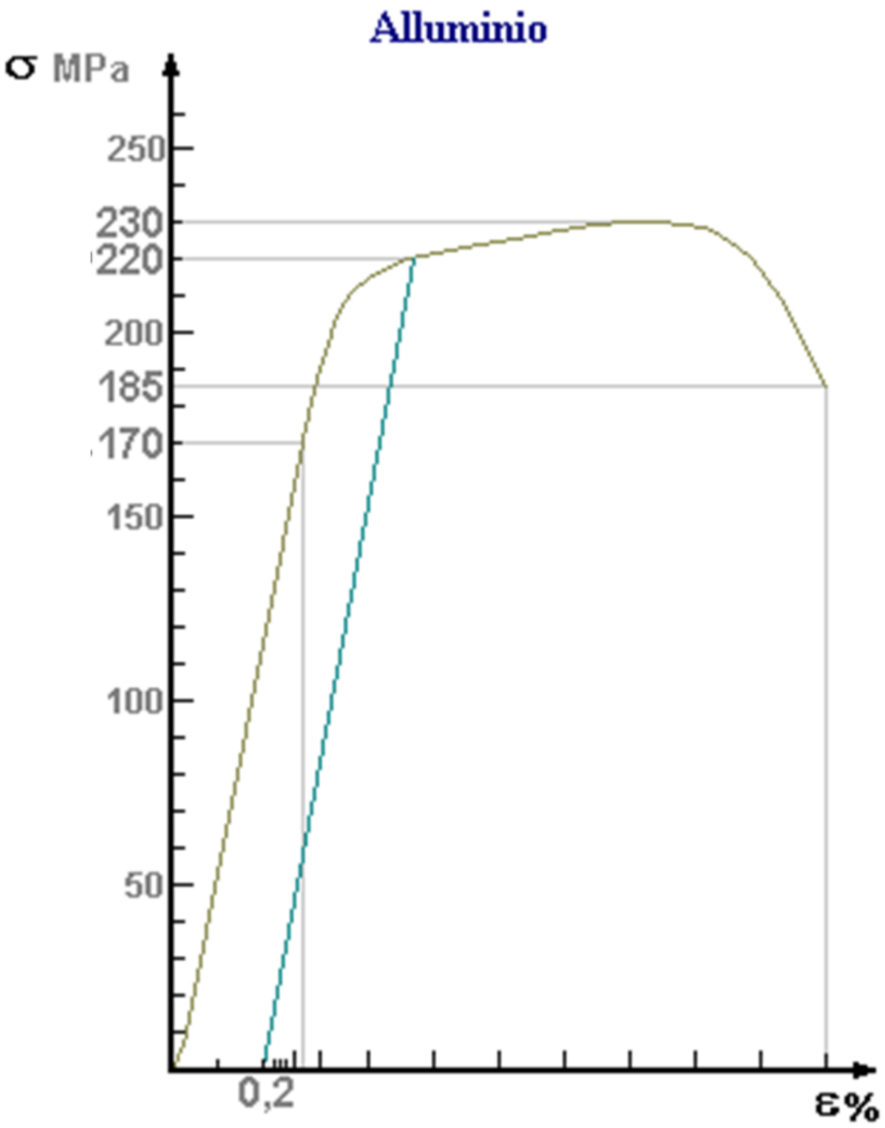
The σ-ε curve in figure comes from the tensile test of an aluminum specimen. Select the right sentence (only one):

a) The specimen had only elastic deformation

a) The specimen underwent necking

a) The yield strenght was not reached in the test

Quiz 3



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